

VASVI JAIN

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EDUCATION

M.S. Biology, New York University	2025 – 2026
B.S. Neural Science & Computer Science, New York University	2021 – 2025

RESEARCH EXPERIENCE

Research Assistant — Tsien Lab, NYU *PI: Dr. Richard Tsien* May 2023 – Present

- **Master's thesis:** investigating cortico-striatal synaptic dysfunction in a Timothy Syndrome Type 2 mouse model, connecting calcium-channel dysfunction to activity-dependent pathways (MAPK/ERK signaling, MeCP2 regulation, c-Fos expression).
- Perform RNAscope, immunohistochemistry, cryostat sectioning, confocal microscopy, and western blotting on cortical and striatal tissue, with colocalization analysis of confocal images.
- **Undergraduate senior thesis:** characterized the interaction between the Cav1.2 β -subunit and CaMKII and its role in CaMKII translocation during long-term potentiation (LTP).
- Designed and executed molecular cloning, transfection, and FRET/FLIM and phosphorylation assays in HEK293T cells and primary rat cortical neurons to probe mechanisms of synaptic plasticity, and analyzed FRET/FLIM lifetime data.

Neuroscience Research Intern — Eikon Therapeutics May 2026 – August 2026

- Support translational neuroscience research using human iPSC-derived neuronal models relevant to Parkinson's disease in a drug-discovery setting.
- Maintain and characterize stable neuronal cell lines, including validation of neuronal marker expression, and manage reagent stocks and materials.
- Perform qPCR, Incucyte live-cell imaging, and imaging-based assay workflows; troubleshoot and refine protocols to improve reproducibility; qualitatively analyzed immunocytochemistry (ICC) imaging data.
- Contribute to assay development and reproducibility testing in 384-well plate formats for high-throughput screening applications.

Research Assistant (Rotation) — Klann Lab, NYU *PI: Dr. Eric Klann* Sep 2025 – May 2026

- Completed a rotation preparing ribosome-associated complexes for downstream cryo-EM analysis; maintained hiPSCs and differentiated NGN2-induced neurons.
- Performed polysome profiling and RNase-based workflows to enrich ribosomal complexes, and analyzed the resulting polysome profiling data.
- Applied biochemical approaches (co-immunoprecipitation, western blotting) to assess possible ribosome-tau interactions.

Research Assistant — Suzuki Lab, NYU *PI: Dr. Wendy Suzuki* Jan 2023 – Sep 2023

- Recruited and scheduled adult participants; conducted study sessions and coded research data.

Research Assistant — OLab, NYU *PI: Dr. Eric Oermann* May 2023 – Sep 2023

- Contributed data analysis and code to a bibliometric study of pediatric spine surgery (published, Neurosurgery Practice 2024).

Research Assistant — Lab for the Developing Mind, NYU *PI: Dr. Moira Dillon* Sep 2022 – Dec 2022

- Recruited participants and coded research data for developmental cognition studies.

TECHNICAL SKILLS

Molecular biology: PCR / qPCR, Molecular cloning, Bacterial transformation, Transfection, Polysome profiling, Nucleic acid extraction & quantification, AAV and Lentivirus transductions

Biochemistry: Western blotting, Co-Immunoprecipitation, RAPPL, RNAscope, Immunohistochemistry, Immunocytochemistry

Cell culture: HEK293T, hiPSCs, NGN2-iNeurons, Mouse astrocyte culture; Stable line maintenance, Aseptic technique

Imaging & analysis: Confocal, brightfield, FLIM/FRET, Incucyte; Python, pandas, NumPy, scikit-learn, matplotlib

PUBLICATIONS

Mir, J.M.; Kurland, D.B.; Cheung, A.T.M.; Liu, A.; Shlobin, N.A.; Alber, D.; Rai, S.; Jain, V.; Rodriguez-Olaverri, J.C.; Anderson, R.C.E.; Lau, D.; Kondziolka, D.; Oermann, E.K. "The Evolution of Pediatric Spine Surgery: A Bibliometric Analysis of Publications From 1902 to 2023." Neurosurgery Practice (2024).
Evans, H.T.; Kalavai, S.V.; Wu, V.; Polavarapu, A.; Balamoti, E.; Liu, W.J.; Mosto, O.; Alder, D.; Jain, V.; Denil, E.; Yu, A.; Assenso, E.; Oliveria, M.; Moniz, T.; Golhan, E.N.; Zhang, X.; Sheehan, C.J.; Klann, E. "Tau-induced ribosomal collisions impair memory through the activation of the integrated stress response." bioRxiv (2026).

AWARDS & PRESENTATIONS

- Dean's Undergraduate Research Fund (DURF), NYU — 2022 and 2024.
- Poster presentations: NYU CNS Neural Science Festival; NYU Undergraduate Research Conference.